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Genetic Improvement of US Soybean in Maturity Groups V, VI, and VII

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ABSTRACT

Soybean [*Glycine max* (L.) Merr.] is an important source of protein and vegetable oil. Genetic improvement of soybean seed yield and composition are ultimate breeding goals. During the past 80 yr, breeders have selected for high yield and other desired traits to make genetic improvements. To quantify the genetic changes to seed yield, yield stability, and other important agronomic and end-use quality traits, we evaluated 93 soybean cultivars in Maturity Groups (MG) V, VI, and VII that were released from 1928 to 2008. Replicated yield trials specific for each MG set of cultivars were conducted during 2010 to 2011 in a total of 27 southeastern US year-location environments. A mixed linear model was used to calculate best linear unbiased predictors (BLUPs) for each cultivar for each measured trait within each MG. Regressed cultivar effect BLUPs of each trait by year of cultivar release revealed positive linear trends for annualized genetic yield gains of 17.6, 13.5, and 10.3 kg ha⁻¹ yr⁻¹ for MG V, VI, and VII, respectively. Averaged across MGs, the annualized rate of genetic gain was estimated to be 13.7 kg ha⁻¹ yr⁻¹. Yield stability analyses revealed significant differences in regression coefficients (*b*), which were >1.0 for newer cultivars but <1.0 for older cultivars. Overall, the average annualized rate of genetic gains for seed yield reported herein are equivalent to those previously reported, indicating that a yield plateau has not been reached for MG V, VI or VII soybean cultivars.

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Abbreviations: BLUP, best linear unbiased predictor; MG, maturity group.

SOYBEAN [*Glycine max* (L.) Merr] is the world's largest source of animal protein feed and the second largest source of vegetable oil (Wilson, 2008). At the present time, it is forecasted that consumption is rising faster than production (USDA, 2017; USDA-NASS, 2018). Soybean, along with corn (*Zea mays* L.), wheat (*Triticum aestivum* L.), and rice (*Oryza sativa* L.), provides roughly two-thirds of the total caloric content of food nutrition to humans globally (Zhao et al., 2017). A recent estimate of the annual noncompounded linear percentage (1.3%) increase in global soybean yield indicated that this yield gain relative to population growth rate is falling short of demand (Egli, 2008; Ray et al., 2013).

The United States currently produces ~50% of the world's soybean (<https://apps.fas.usda.gov/>) and average on-farm

Published in Crop Sci. 59:1838–1852 (2019).
doi: 10.2135/cropsci2018.10.0627

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commercial soybean yields have steadily risen during the past 80 yr because of concerted efforts of both public and private plant breeders that have developed and released new cultivars with improved yield potential, desirable agronomic traits, and disease resistance (Carter et al., 2004). To better understand the yield potential of soybean in the United States and to assess the current rate of progress, soybean breeders have periodically conducted studies to measure one of the primary factors under their selective control, which is the rate of genetic gain in yield made across time through the release of breeder-developed improved soybean cultivars.

Yield potential for agricultural crops was defined by Evans and Fischer (1999) as the yield of a cultivar when grown in environments to which it is adapted, with water and nutrients nonlimiting and with abiotic and biotic stresses effectively controlled. The definition proposed by Loomis and Amthor (1999) implies that achieving high yields to meet yield potential is a relatively straightforward process so long as a minimal fitness cost is associated with the optimal partitioning of new assimilates, arising from photosynthesis, to the plant's roots, stem, leaves, and reproductive structures. In evaluating the relationship between yield potential and photosynthesis, Koester et al. (2014) attributed seed yield increases in modern cultivars compared with older cultivars to physiological enhancements in the plant canopy structure, which served to increase the efficiencies for light interception, light energy converted into biomass, and the partitioning of biomass into seed. Hence, it was concluded that the calculated radiation use efficiency for soybean was still below the theoretical maximum for the crop, indicating that the yield potential could be raised further. The two definitions of yield potential, described above, lie in direct contrast with yielding capacity, which describes the cultivation of crops in perfectly flawless growing conditions, completely free of abiotic and biotic stresses (Bingham, 1967; MacKey, 1979; Evans and Fischer, 1999; Specht et al., 1999). On an annual basis, soybean breeders aim to increase the genetic gain of the crop by recurrently selecting plants in targeted production environments that exhibit desirable agronomic traits, such as high yield, resistance to lodging, and tolerance of prevailing abiotic and biotic environmental stressors (Evans and Fischer, 1999). Given the many fluctuating abiotic and biotic environmental stressors that annually limit soybean yield, it was conjectured that a biological limit of 8000 kg ha⁻¹ for soybean yield improvement could be the yield maxima for the crop (Specht et al., 1999).

The arrival of new technologies across time has aided soybean breeders in steadily increasing genetic gains. Specht et al. (1999) postulated that the rapid adoption of genetic and agronomic technologies by soybean breeders and producers was a major attributor in the annual

improvement and gains associated with US soybean yields from 1924 to 1998. Some examples of past and recent genetic and agronomic technological advances that have aided both breeders and producers in achieving high yields are molecular breeding (Collard et al., 2005), precision agriculture (Pierce and Nowak, 1999), smart farming (Trippichio et al., 2015), and good agronomic practices (Pittelcow et al., 2015).

Another factor implicated in raising the yield of soybean across time is the carbon fertilization effect stemming from the annual rise of atmospheric CO₂ levels, which were estimated at 1.5 μL L⁻¹ yr⁻¹ from 1924 to 1997 (Allen et al., 1998). The mean annual concentration of atmospheric CO₂ levels is now 50 moles per million mole of dry air more concentrated than when measured three decades ago (Jenkinson et al., 1991; Sakurai et al., 2014; Specht et al., 2014). Elevated CO₂ levels have benefited crop yield by improving plant photosynthesis, water use efficiency, and biomass accumulation (Kimball, 1983; Waggoner, 1984; Allen et al., 1987; Roche, 2015; Keep et al., 2016). Allen et al. (1987) predicted that a doubling of atmospheric CO₂ levels from 1958 to 2058 could potentially increase soybean yields by as much as 32%. Similarly, Specht et al. (1999) estimated that ~15% of the 31.4 kg ha⁻¹ rate of on-farm US soybean yield improvement from 1972 to 1997 was a result of increasing atmospheric CO₂ levels. More recently, Sakurai et al. (2014) determined that soybean yields were boosted 5.8% on average from 2002 to 2006 because of increases in atmospheric CO₂ compared with levels quantified in 1980 and reported that the C fertilization effect was positively correlated with solar radiation during the growing period.

It has also been found that physiological enhancements to the soybean plant across time have boosted yields. Some studies have noted that the transpiration rates of modern cultivars are higher than those of older cultivars, which may have inadvertently resulted from high selection pressure for yield (Roche, 2015; Keep et al., 2016). Additionally, physiological differences in the canopy structure were also noticed between modern cultivars and landraces, which are thought to boost yields by enhancing photosynthesis (Koester et al., 2014). Given that the world record for soybean yield was awarded in 2016 to a Georgia producer, who achieved a record yield of 11,500 kg ha⁻¹ (<http://www.cornandsoybeandigest.com>), it is evident that the biological limit to soybean yield improvement is much greater than previously speculated, which highlights the importance of maximizing genetic gains by developing cultivars with increased photosynthetic ability, water use efficiency, and yield potential.

Genetic yield gains are estimated by simultaneously evaluating currently and previously released cultivars from past decades in one or more of their adapted field production environments in randomized and replicated

plot yield trials. After harvest, yield data are then analyzed using linear regression analyses to determine if significant yield differences are present between new and old cultivars that were released across years within adapted regions (Cox et al., 1988; Slafer, 1994; Wilcox, 2001; Specht et al., 2014). In this manner, direct estimates of yield can be compared between historical and modern-day cultivars to calculate annualized rates of genetic gain across time (Evans, 1980). A list of previously conducted studies that estimated genetic yield gains for cultivars in designated North American soybean maturity groups (MGs) is given in Table 1, which allows a comparison of genetic gains made across time among cultivars across MGs.

The first study estimating genetic yield gains for soybean cultivars from MGs VI, VII, and VIII that were released during the 1940s, 1950s, 1960s, and 1970s was conducted by Boerma (1979). The author evaluated six cultivars from each MG across 3 yr in a single Georgia environment and reported genetic yield gains of 13.7 kg ha⁻¹ by regressing seed yield on year of release, which equated to a 0.7% yr⁻¹ increase when averaged across all MGs. After evaluating 18 MG VI, VII, and VIII soybean cultivars released between 1945 and 1983 across 3 yr in four Florida environments, Salado-Navarro et al. (1993) reported genetic yield gains ranging from 15.9 to 18.7 kg ha⁻¹ yr⁻¹ that were slightly higher than those reported by Boerma (1979). Ustun et al. (2001) conducted a similar trial in four Tennessee environments across 3 yr using a small sample size of eight MG V, VI, and VII cultivars released from 1954 to 1987 and reported genetic yield gains of 14.4 kg ha⁻¹ yr⁻¹, pointing out that a yield plateau had not been reached in soybean, as had been seen in similar genetic gain studies for wheat (Schmidt, 1984; Schmidt and Worrall, 1984; Waddington et al., 1986). Additionally, Rogers et al. (2015) evaluated 45 southern

MG IV to VI cultivars released from 1928 to 2008 in three experiments from 2007 to 2008 and observed linear yield increases of 16.8 kg ha⁻¹ yr⁻¹ for all MGs combined, with genetic gains estimated at 16.4 kg ha⁻¹ yr⁻¹ for MG V cultivars and 12.4 kg ha⁻¹ yr⁻¹ for MG VI cultivars.

When genetic gain studies are conducted, it is common to simultaneously evaluate yield stability (Voldeng et al., 1997; Wilcox et al., 1979; De Bruin and Pedersen, 2008). Yield stability analyses are often used to better understand genotype × environment interactions and provide the breeders with estimates of adaptation of cultivars and breeding lines evaluated in yield trials across multiple environments. There are two types of stability: (i) static stability, in which the performance of a genotype remains relatively unchanged irrespective of environment, and (ii) dynamic stability, in which the performance of a genotype changes in a predictable manner when evaluated in a wide range of environments (Becker and Léon, 1988; Tollenaar and Lee, 2002). Static stability is also regarded as the Type 1 stability, whereas dynamic stability is often ascribed to the Type 2 stability concept (Lin et al., 1986; Becker and Léon, 1988). Herein, the Finlay and Wilkinson method was used for calculating yield stability coefficients for each genotype, which is a measure of the Type 2 dynamic stability concept, where the environments are expected to influence cultivar yields (Finlay and Wilkinson, 1963; Tollenaar and Lee, 2002). Recently, Xavier et al. (2018) conducted a genome-wide association study and used the same Finlay and Wilkinson approach to provide performance stability estimates for genotypes in a large soybean nested association mapping population, revealing a genomic region associated with yield stability on chromosome 18.

Selection for yield across time can have a positive or negative indirect selection impact on other agronomic traits, such as lodging, plant height, days to maturity,

Table 1. Summary of published studies to estimate the annualized rate of genetic gain for seed yield in North American soybean for Maturity Groups (MGs) 000 through VIII.

MGs	Cultivar release dates	Genetic gain estimate kg ha ⁻¹ yr ⁻¹	Citation
000, 00, 0	1934–1992	11.0	Voldeng et al. (1997)
00, 0, I, II, III, IV	1902–1977	19.0	Specht and Williams (1984)
00, 0, I, II, III, IV	1939–1999	22.0–31.0	Wilcox (2001)
I, II, III, IV	1929–1976	16.0	Luedders (1977)
I, II, III	1938–2004	25.0	De Bruin and Pedersen (2008)
II, III	1923–1974	11.0–13.0	Wilcox et al. (1979)
III	1923–2007	26.5	Koester et al. (2014)
II, III, IV	1923–2008	19.5–23.1	Rincker et al. (2014)
III, IV, V	1980–2015	43.0	de Felipe et al. (2016)†
IV, V, VI	1928–2008	12.4–21.7	Rogers et al. (2015)
V, VI, VII	1942–1987	14.0	Ustun et al. (2001)
VI, VII, VIII	1942–1973	14.0	Boerma (1979)
VI, VII, VIII	1945–1983	16.0–19.0	Salado-Navarro et al. (1993)
00–VIII	1928–2008	23.0	Specht et al. (2014)‡

† Study conducted in South America (Argentina) using US commercial cultivars.

‡ Study encompassed all continental US MGs.

and seed size. Moreover, breeders may not have solely selected for yield but may have simultaneously selected for one or more other agronomic traits when making advancement decisions. Many of the genetic gain studies (Table 1) reported an improvement in lodging resistance by as much as one unit when using a standard 1-to-5 rating scale to assess lodging in yield trials (Luedders, 1977; Wilcox et al., 1979; Specht and Williams, 1984; Voldeng et al., 1997; de Felipe et al., 2016). Boerma (1979) observed no changes in plant height across 60 yr between old and new cultivars in MG VI and MG VII but did note plant height reductions among MG VIII cultivars. Ustun et al. (2001) noted a trend where plant height decreased for MG V and VI cultivars between 1940 and 1970 but then increased from 1970 to 2000. A similar decrease in plant height was reported by Wilcox (2001) and Rincker et al. (2014) for MG II, III, and IV cultivars across six decades and eight decades, respectively. Generally, plant height and lodging are thought to be interrelated, thus it is logical that selection for resistance to lodging may have indirectly led to a reduction in plant height (Ustun et al., 2001). Ustun et al. (2001) also observed that maturity was shifting towards earlier-maturing genotypes for MG V, VI, and VII cultivars, although the total combined sample size evaluated was small ($n = 8$). This maturity observation was also documented by de Felipe et al. (2016) in a genetic gain study conducted in Argentina using a set of 55 US MG V cultivars.

Selection for yield across time may also have a positive or negative indirect selection impact on end-use quality traits. For example, it is widely reported that protein and yield, and protein and oil share an inverse relationship (Burton, 1984; Wilcox and Cavins, 1995; Cober and Voldeng, 1999), so it was not surprising that repetitive breeder selection for increased yield across time has led to a reduction in seed protein and an increase in oil for many MGs in many studies (Wilcox et al., 1979; Voldeng et al., 1997; Ustun et al., 2001; Rincker et al., 2014; de Felipe et al., 2016). Ustun et al. (2001) noted significant differences in protein content between old and new MG V to VII cultivars, but the slope of protein change across time between old and new cultivars was not significant. Rogers et al. (2015) observed a significant $0.17 \text{ g kg}^{-1} \text{ yr}^{-1}$ linear decrease in seed protein concentration between old and new MG IV to VI cultivars released between 1928 and 2008, which amounted to a $-0.32 \text{ g kg}^{-1} \text{ yr}^{-1}$ decline for MGs V and VI. Rincker et al. (2014) observed no significant changes for seed weight and quality among MG II, III, and IV cultivars across eight decades.

The objectives of this study were (i) to estimate genetic gain for yield and other agronomic and end-use quality traits for US adapted MG V, VI, and VII soybean cultivars released from 1928 to 2008, and (ii) to compare

the yield stabilities among cultivars within each of MG V, VI, and VII.

MATERIALS AND METHODS

Plant Materials

A total of 93 soybean cultivars belonging to MG V, VI, and VII, which were released by both public and private soybean breeders from 1928 to 2008, were selected for evaluation (Table 2). Of these cultivars, 36 were classified as MG V, 30 as MG VI, and 27 as MG VII. The cultivars were selected on the basis of their widespread adoption by growers after their release, provided they were included as entries in past regional and state-wide yield trials, which are annually conducted across the southern United States within each designated MG. Of the 93 entries, 50 represented commercial cultivars that were developed by Pioneer, Syngenta, Monsanto, or the now defunct Coker Pedigreed Seed (Hartsville, SC) and McNair Seed (Laurinburg, NC) Companies, equating to 22 of 36 cultivars for MG V, 16 of 30 entries for MG VI, and 12 of 27 entries for MG VII (Table 2). Proprietary seed from the private companies was kindly provided by representatives of their respective companies, except for cultivars developed by the Coker and McNair Seed Companies, which were obtained from the USDA Soybean Germplasm Collection (Urbana, IL). Seed of public cultivars were procured from the respective breeding programs. The PI numbers for the cultivars were obtained from Germplasm Resources Information Network (GRIN) (<https://www.ars-grin.gov/>), and the year of release for each cultivar and pedigrees (if listed) were either acquired from published plant registration articles or provided by the company granting the seed. Seed for all of the cultivars were increased in one field at the University of Georgia's Plant Sciences Farm, Athens, GA, the year prior to the first yield tests.

Yield Trials and Measured Traits

Cultivars within each MG were subjected to performance trials conducted in 2010 and 2011 at state variety testing locations held for each MG. There were 27 year-location environments, which amounted to 12 year-location environments for MG V, six year-location environments for MG VI, and nine year-location environments for MG VII (Fig. 1B). Each MG cultivar set was treated as a separate experiment conducted as a randomized complete block design with either three (MG V) or four (MG VI and VII) replications. Approximately 26 seeds m^{-1} of each cultivar were planted in four-row yield plots with 0.76-m spacing between rows; plot lengths ranged from 2.9 to 6.1 m, depending on test location. Seed was not treated with insecticides and fungicides prior to planting, and standard yield trial testing protocols were followed at each location. Seed yield calculations for each cultivar were based on harvesting the middle two rows of each four-row plot with a plot combine; yield was adjusted to 130 g kg^{-1} moisture. Lodging score was assigned on a scale of 1 (erect) to 5 (prostrate) at maturity. At maturity, plant height was measured as the average of three plants from the ground to the terminal node. The date of plant maturity was recorded when 95% of the pods

Table 2. Soybean cultivars released from 1928 to 2008 selected for genetic gain study from Maturity Group (MG) V through VII. Cultivars are ordered by year of release within each MG.

MG	Decade of release	Cultivar	Accession no.	Year of release	Breeding program†	Pedigree
V	1940–1950	S-100	PI 548488	1942	P	Selection from Illini
	1950–1960	Dorman	PI 548653	1952	P	Dunfield/Arksoy 2913
	1950–1960	Hill	PI 548654	1959	P	Haberlandt/Dunfield//S-100/CNS
	1960–1970	Dare	PI 548987	1965	P	Hill//Roanoke/Ogden
	1960–1970	York	PI 553038	1967	P	Dorman/Hood
	1970–1980	Essex	PI 548667	1972	P	Lee/S55-7075
	1970–1980	Forrest	PI 548655	1973	P	Dyer/Bragg
	1970–1980	McNair 500	PI 556506	1976	C	Davis/Lee 68
	1970–1980	MON033	NA‡	1981§	C	NA
	1980–1990	9592	NA	1984	C	NA
	1980–1990	Coker 485	PI 556743	1984	C	Centennial/Hampton 266/Bragg//Hutton
	1980–1990	MON043	NA	1986§	C	NA
	1980–1990	Hutcheson	PI 518664	1987	P	V68-1034/Essex
	1980–1990	Coker 6955	PI 556903	1989	C	NA
	1980–1990	MON021	PI 556861	1989	C	NA
	1990–2000	S59-60	NA	1991	C	NA
	1990–2000	Holladay	PI 572239	1993	P	N77-179/Johnston
	1990–2000	Clifford	PI 596414	1994	P	N77-114/N77-179
	1990–2000	MON034	NA	1995§	C	NA
	1990–2000	MON035	NA	1995§	C	NA
	1990–2000	MON036	NA	1995	C	NA
	1990–2000	9594	NA	1996	C	NA
	1990–2000	MON022	NA	1996	C	NA
	1990–2000	Graham	PI 594922	1996	P	N77-114/Pixie
	1990–2000	S59-V6	NA	1997	C	NA
	1990–2000	MON023	NA	1998	C	NA
	2000–2010	MON028	NA	2001	C	NA
	2000–2010	MON040	NA	2001	C	NA
	2000–2010	5601T	PI 630984	2001	P	Hutcheson/TN89-39
	2000–2010	S56-D7	NA	2002	C	NA
	2000–2010	MON027	NA	2003	C	NA
	2000–2010	95M50	NA	2005	C	NA
	2000–2010	Jake	PI 643912	2006	P	S94-1867/Anand
	2000–2010	S52-F2	NA	2007	C	NA
	2000–2010	Osage	PI 648270	2007	P	Hartz 5545/KS4895
	2000–2010	95Y40	NA	2008	C	NA
VI	1930–1940	Arksoy	PI 548438	1928	P	Selection from PI 37335
	1940–1950	Ogden	PI 548477	1940	P	Tokyo/PI 54610
	1950–1960	Lee	PI 548656	1954	P	S-100/CNS
	1950–1960	Hood	PI 548980	1958	P	Roanoke/Ogden/CNS
	1960–1970	Pickett	PI 548988	1965	P	D49–2491/Dorman//Lee/Peking
	1960–1970	Davis	PI 553039	1966	P	Roanoke/N45-745//Ralsoy/Ogden
	1970–1980	Tracy	PI 548983	1973	P	Hill/PI 171442 D60-9647//FC 31745/D49-2510
	1970–1980	Centennial	PI 548975	1976	P	Hill//D49-2491/Jackson//Pickett 71
	1970–1980	Coker 136	PI 628819	1978	C	N59-6800/Co Hampton 266
	1970–1980	Coker156	PI 556564	1979	C	NA
	1980–1990	Lefflore	PI 548981	1984	P	Centennial///Forrest//D68-18/PI 88788
	1980–1990	Young	PI 508266	1984	P	Davis/Essex
	1980–1990	Coker 686	PI 556827	1987	C	NA
	1980–1990	MON026	PI 527704	1987	C	NA
	1990–2000	MON024	PI 556858	1988	C	NA
	1990–2000	Brim	PI 548986	1990	P	Young//Tracy/Ransom
	1990–2000	9692	NA	1992	C	NA
	1990–2000	S65-50	NA	1992	C	NA
	1990–2000	MON025	NA	1995	C	NA

Table 2. Continued.

MG	Decade of release	Cultivar	Accession no.	Year of release	Breeding program†	Pedigree
	1990–2000	Dillon	PI 592756	1995	P	Centennial/Young
	2000–2010	S61-Q2	NA	1996	C	NA
	1990–2000	Boggs	PI 602597	1997	P	G81-152/Coker 6738
	2000–2010	Musen	PI 599333	1997	P	Hutcheson/LeFlore
	1990–2000	96B21	NA	1999	C	NA
	2000–2010	NC Roy	PI 617045	2001	P	Holladay/Brim
	2000–2010	MON032	NA	2003	C	NA
	2000–2010	96M20	NA	2004	C	NA
	2000–2010	MON030	NA	2005	C	NA
	2000–2010	S65-M3	NA	2005	C	NA
	1990–2000	S62-66	NA	2007	C	NA
VII	1940–1950	CNS	PI 548445	1927	P	Selection from Clemson
	1940–1950	Roanoke	PI 548485	1946	P	Selection from Nanking
	1950–1960	Jackson	PI 548657	1953	P	Volstate/Palmetto
	1960–1970	Bragg	PI 548660	1963	P	Jackson//S-100/CNS
	1970–1980	Ransom	PI 548989	1970	P	Perry/Lee
	1970–1980	GaSoy17	PI 553046	1977	P	Bragg/Hood
	1970–1980	Coker 237	PI 556536	1978	C	NA
	1970–1980	Braxton	PI 548659	1979	P	Jackson//S-100/CNS///Bragg/D60-7965
	1980–1990	Coker 317	PI 556623	1982	C	NA
	1980–1990	MON039	NA	1982§	C	NA
	1980–1990	Coker 6847	PI 556862	1988	C	NA
	1980–1990	Thomas	PI 522236	1988	P	Centennial/F71-1138
	1980–1990	Stonewall	PI 531068	1989	P	D68-216/Ransom////Centennial///Forrest//Cobb/D68-216
	1990–2000	Hagood	PI 555453	1990	P	Centennial/Young
	1990–2000	Cook	PI 553045	1991	P	Braxton/Young
	1990–2000	S75-55	NA	1993	C	NA
	1990–2000	Benning	PI 595645	1996	P	Hutcheson/Coker 6738
	1990–2000	S73-Z5	NA	1997	C	NA
	1990–2000	MON044	NA	1999	C	NA
	2000–2010	97B52	NA	2001	C	NA
	2000–2010	S76-L9	NA	2002	C	NA
	2000–2010	NC-Raleigh	PI 641156	2002	P	N85-492/N88-480
	2000–2010	Woodruff	NA	2005	P	Boggs/N7001
	2000–2010	97M50	NA	2006	C	NA
	2000–2010	N7002	PI 647085	2007	P	N7001/Cook
	2000–2010	MON031	NA	2008	C	NA
	2000–2010	S74-W6	NA	2008	C	NA

† P denotes public breeding program and C denotes commercial breeding program.

‡ NA, not available due to company policy.

§ Denotes the first year cultivar was found in variety testing reports.

exhibited a mature pod color at the R8 growth stage (Fehr et al., 1971). The difference in days between the maturity date and the planting date represented days to maturity. Seed weight was recorded by weighing a random sample of 100 harvested seeds, which were taken from each field plot from first two replications or three replications. Seed quality was visually scored using a 1-to-5 scale typically used in USDA Northern and Southern Uniform Soybean Tests, with a score of 1 indicating very good quality and a score of 5 indicating poor quality (Crochet and Hughes, 2012). Random seed samples from harvested plots were analyzed for protein and oil concentrations using near-infrared reflectance (NIR) (Pertin Instruments, Model DA 7250) and quantified levels were expressed on a 130 g kg⁻¹ moisture basis. Agronomic

traits and seed composition were collected from some of the environments each year, depending on the MGs (Supplemental Table S1).

Statistical Analyses

To estimate genetic gains made for yield and the other traits across 80 yr within each MG, a mixed linear model was used in SAS 9.4 (PROC Mixed; SAS Institute, 2013) to calculate best linear unbiased predictors (BLUPs) for each measured trait for each cultivar within each MG, with environment and replication nested within environment treated as fixed effects and cultivar and cultivar × environment interaction treated as random effects. Subsequently, BLUPs of the cultivar effects for each trait were regressed on the cultivar release year using

PROC REG in SAS 9.4 (SAS Institute, 2013) to estimate annualized rates of genetic change across years within each MG for each trait. The regression of yield vs. time allows for estimates of absolute gain rates in kilograms per hectare per year (Egli, 2008). The yield stability of each cultivar was first estimated using the Finlay–Wilkinson method by calculating the regression coefficient (b) for each cultivar by regressing the mean yield of each cultivar in a given environment on the mean yield of all cultivars in that environment. The stability of a given cultivar is considered greater than average if the regression coefficient is <1.0 and less than average if the regression coefficient is >1.0 ; a stability = 1.0 is considered average (Tollenaar and Lee, 2002). In this manner, the yield stability or adaptive ability of cultivars could be assessed by comparing the regression coefficients. Subsequently, the computed regression coefficient (b) for each cultivar in a given MG was then regressed on year of cultivar release to compare the cultivar regression coefficients across time. Pearson product–moment pairwise correlations among trait means of cultivars within each MG were calculated using JMP Pro 14.1.0 (SAS Institute).

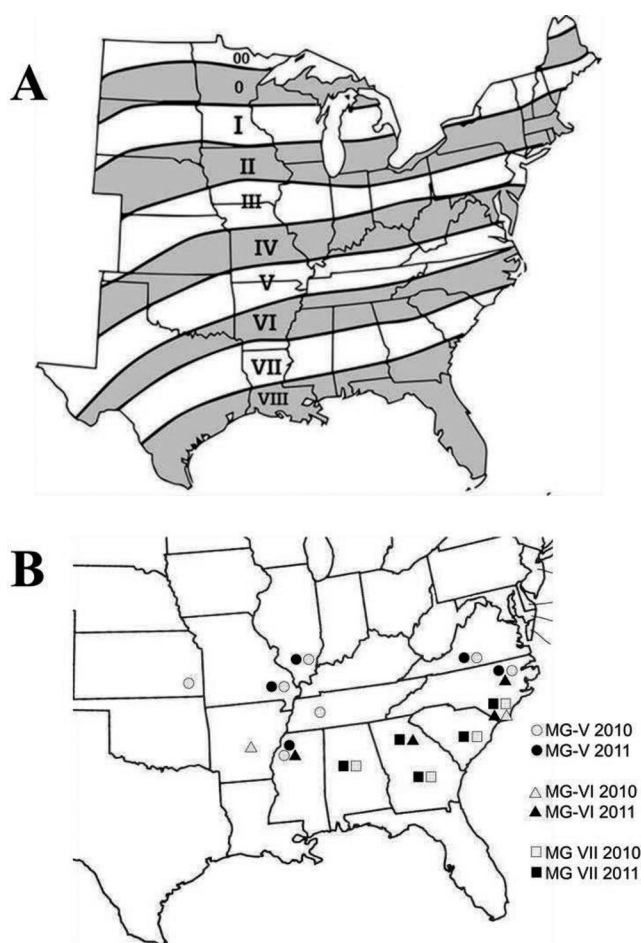


Fig. 1. Soybean maturity groups and testing locations. (A) Map depicting approximate growing regions of US defined soybean maturity groups (MGs) (Stowe and Dunphy, 2017). (B) Geographical locations of the 27 total southern US year-location environments used to conduct yield trials in 2010 to 2011 for soybean cultivars selected from MG V to VII, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII.

RESULTS AND DISCUSSION

Genetic Gain for Agronomic Traits

Both public and private sector soybean breeders developed and released soybean cultivars from 1928 to 2008, by primarily selecting for high yield across eight decades. That effort has generated significant genetic gains for seed yield in MG V, VI, and VII (Table 3, Fig. 2). The annualized linear rates of genetic yield improvement were estimated at 17.6 kg ha⁻¹ yr⁻¹ for MG V (Fig. 2A), 13.5 kg ha⁻¹ yr⁻¹ for MG VI (Fig. 2B), and 10.3 kg ha⁻¹ yr⁻¹ for MG VII (Fig. 2C). The decline in genetic gain rate from MG V to VII is similar to the decline in genetic gain rates from MG II to IV reported by Rincker et al. (2014); for

Table 3. Estimated rates of genetic gain of five agronomic and three end-use quality traits for selected soybean cultivars released from 1928 to 2008: (A) Maturity Group (MG) V ($n = 36$), (B) MG VI ($n = 30$) and (C) MG VII ($n = 27$).

Trait	MG	b	$\pm$ SE	R^2
Agronomic traits†				
Yield (kg ha ⁻¹ yr ⁻¹)	V	17.537***	1.307	0.841
	VI	13.503***	1.534	0.734
	VII	10.282***	1.139	0.765
Lodging (score yr ⁻¹ ‡)	V	-0.011***	0.003	0.282
	VI	-0.009***	0.002	0.252
	VII	-0.013***	0.002	0.554
Plant height (cm yr ⁻¹)	V	-0.118NS§	0.102	0.037
	VI	-0.032NS	0.044	0.019
	VII	-0.061NS	0.061	0.037
Days to maturity (d yr ⁻¹ ¶)	V	0.054NS	0.030	0.084
	VI	-0.002NS	0.033	0.001
	VII	-0.022NS	0.019	0.050
100-seed weight (g yr ⁻¹)	V	0.009NS	0.014	0.009
	VI	-0.019NS	0.016	0.049
	VII	-0.012NS	0.008	0.074
End-use quality traits				
Seed protein (g kg ⁻¹ yr ⁻¹ ‡)	V	0.022NS	0.095	0.002
	VI	-0.234**	0.070	0.270
	VII	-0.092NS	0.052	0.110
Seed oil (g kg ⁻¹ yr ⁻¹ ‡)	V	-0.026NS	0.072	0.004
	VI	0.074NS	0.059	0.053
	VII	0.041NS	0.057	0.021
Seed quality (score yr ⁻¹ ††)	V	-0.005***	0.001	0.343
	VI	-0.001NS	0.001	0.003
	VII	-0.001NS	0.001	0.126

, * Significant at the 0.01 and 0.001, probability levels, respectively.

† Estimates were based on best linear unbiased predictor (BLUP) trait values generated from replicated yield trials conducted in 27 total southern US environments from 2010 to 2011. The BLUP trait value of each cultivar was regressed on the cultivar release date to estimate annualized genetic gains made over eight decades for each trait within each MG.

‡ Lodging score is assigned visually from 1 (all plants erect) to 5 (all plants prostrate to the ground).

§ NS, not significant.

¶ Days to maturity start on 1 September with positive BLUP values indicating later maturing cultivars and negative values indicating earlier maturing cultivars.

Protein and oil concentrations are expressed on a 130 g kg⁻¹ moisture basis.

†† Seed quality is assigned visually from 1 (excellent) to 5 (poor).

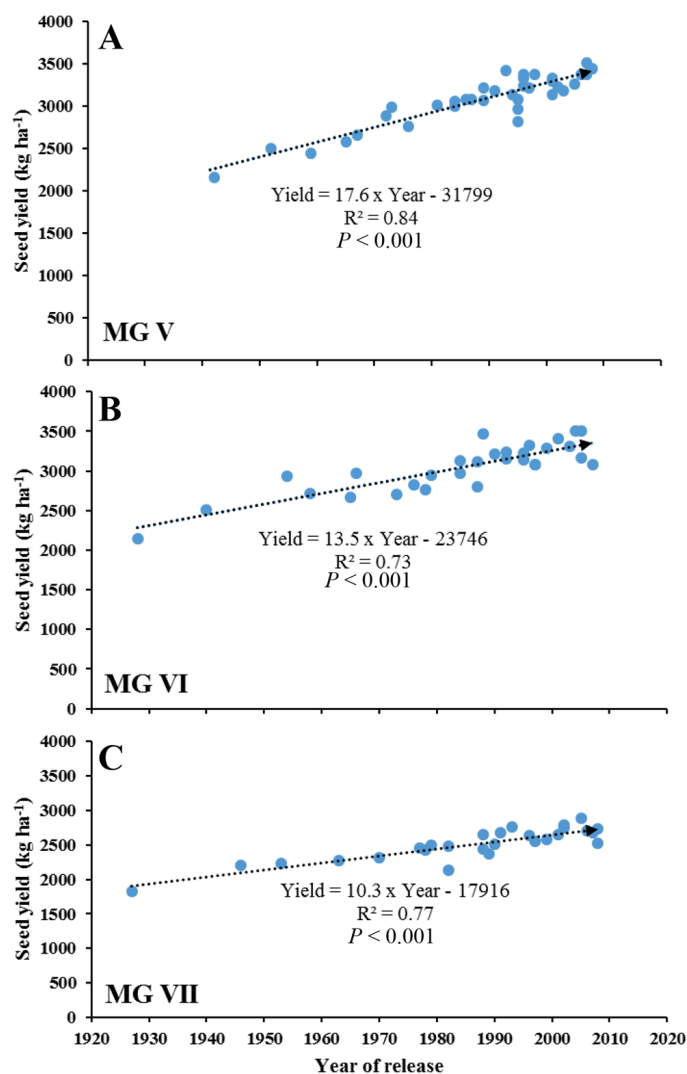


Fig. 2. Scatterplots and regressions depicting genetic yield gains estimated for soybean cultivars in Maturity Group (MG) V through VII released from 1928 to 2008: (A) MG V ($n = 36$), (B) MG VI ($n = 30$), and (C) MG VII ($n = 27$). Estimates were based on best linear unbiased predictor (BLUP) trait values generated from replicated yield trials conducted in 27 total southern US year-location environments in 2010 to 2011, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII. The BLUP trait value for seed yield of each cultivar was regressed on the year of cultivar release to estimate annualized genetic yield gains made over eight decades for each MG.

MG II, the rate was $23.1 \text{ kg ha}^{-1} \text{ yr}^{-1}$, for MG III it was $22.8 \text{ kg ha}^{-1} \text{ yr}^{-1}$, and for MG IV it was $19.5 \text{ kg ha}^{-1} \text{ yr}^{-1}$ (Table 1). This decline is not surprising given that the measured rate of genetic gain is highly dependent on the productivity of the test environments in which that gain is estimated (Rincker et al., 2014). The geographic distribution of the 27 year-location test environments evaluated during 2010 to 2011 is displayed in Fig. 1B. By MG, the average yield was 3078 kg ha^{-1} in the 12 MG V year-location environments, 3048 kg ha^{-1} in the six MG VI year-location environments, and 2507 kg ha^{-1} in the nine MG VII year-location environments. Averaged across

MGs V, VI, and VII, the annualized rate of genetic yield improvement for the 93 selected cultivars was $13.7 \text{ kg ha}^{-1} \text{ yr}^{-1}$ (Fig. 3), the exact same rate of genetic yield gain estimated by Boerma (1979) for a group of 17 selected MG VI, VII, and VIII soybean cultivars. This result was also similar to that estimated by Ustun et al. (2001) for a small sample ($n = 8$) of MG V, VI, and VII soybean cultivars ($14.4 \text{ kg ha}^{-1} \text{ yr}^{-1}$), by Salado-Navarro et al. (1993) for a slightly larger sample ($n = 18$) of MG VI, VII, and VIII soybean cultivars ($15.9\text{--}18.7 \text{ kg ha}^{-1} \text{ yr}^{-1}$), and by Rogers et al. (2015) for a sample of 45 MG IV to VI cultivars ($16.8 \text{ kg ha}^{-1} \text{ yr}^{-1}$). The on-farm yield improvement from 1924 to 2018 was also calculated for the 13 US states that grow MG V to VII soybean, which are Alabama, Arkansas, Georgia, Kentucky, Louisiana, Mississippi, Missouri, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, and Virginia (Supplemental Fig. S1A). The on-farm soybean yield in these states quintupled from 571.6 kg ha^{-1} in 1924 to $2844.7 \text{ kg ha}^{-1}$ in 2018, and a linear regression model fit to the USDA National Agricultural Statistics Service-estimated yield data for these states across the same time period generated a rate of increase of $22.1 \text{ kg ha}^{-1} \text{ yr}^{-1}$ ($R^2 = 0.90$) (Supplemental Fig. S1B). This estimate is consistent with the $23.0 \text{ kg ha}^{-1} \text{ yr}^{-1}$ rate of on-farm soybean yield improvement that was estimated for the continental United States from 1924 to 2012 (Specht et al., 2014). On-farm yield improvement for the 13 states (Supplemental Fig. S1A) was much higher than the genetic gains estimated for MG V, VI, and VII, suggesting that agronomic technologies used by producers also played an important role in the annual improvement of US soybean yields from 1924 to 2012.

The average relative growth rate of genetic gain for the combined MGs herein (0.73%) was also comparable with that previously estimated 30 yr ago by Boerma (1979) for MGs VI to VIII (0.70%). Grassini et al. (2013) evaluated six different statistical models for their performance to fit crop yield trends and determined that a linear rate of gain adequately described all yield trends evaluated, providing the best fit 27% of the time. In our regression analyses displayed in Fig. 2, the coefficients of determination for MG V (0.84), MG VI (0.73), and MG VII (0.77) each successfully demonstrate that high proportions of variance in the dependent variable (yield) were “explained” by the independent variable (year of cultivar release) in the regression model (Rao, 1973; Nagelkerke, 1991). Grassini et al. (2013) also estimated that 4 to 18 yr of “flat” yields are needed to detect a statistically significant yield plateau in cereal crops. Given (i) that calculations for both the annualized rate of genetic gain and the relative yield gain growth rates have remained relatively unchanged (i.e., steadily increased) for >30 yr and (ii) that there is no evidence of 4 to 18 yr of “flat” yields in the 80-yr dataset in our study, it is reasonable to suggest that a yield plateau

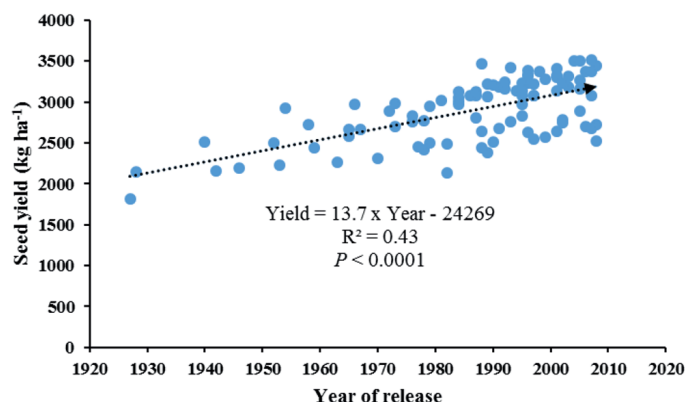


Fig. 3. Scatterplot depicting genetic yield gains estimated for all 93 soybean cultivars selected from Maturity Group (MG) V, VI, and VII and released during 1928 to 2008. Estimates were based on best linear unbiased predictor (BLUP) trait values generated from replicated yield trials conducted in 27 total southern US year-location environments in 2010 to 2011, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII.

has not been reached in the MG V to VII soybean production zones. The same conclusion was reached by Specht et al. (2014) for on-farm US soybean yield from 1924 to 2012, and the same inference can be deduced for on-farm US soybean yield during the same time period for states occupying MGs V to VII in the present study (Supplemental Fig. S1B).

Relative rates of genetic gain were also estimated as a percentage of genetic gain based on yield improvement after the first decade cultivars were released in each MG, which was the 1940s for MG V, the 1930s for MG VI, and the 1940s for MG VII. The resultant rates of genetic gain were 0.9% yr⁻¹ for MG V, 0.7% yr⁻¹ for MG VI, and 0.6% yr⁻¹ for MG VII. Again, these relative genetic gain results for each MG were similar to the 0.7% yr⁻¹ increase previously observed by Boerma (1979) for MG VI, VII, and VIII. Although the current pace of genetic yield improvement for the selected group of MG V, VI, and VII soybean cultivars evaluated herein was maintained during the past three decades, the mean estimated yearly yield gains (0.7% yr⁻¹) are well shy of the projected noncompounded annual linear global increase in soybean yield (1.3%) needed to help feed a global population soon to be ~10 billion people in 40 yr (Ray et al., 2013).

In addition to yield, four other agronomic traits were evaluated: lodging, plant height, days to maturity, and 100-seed weight. Of these four traits, significant differences between old and new soybean cultivars within each MG were only observed for cultivar lodging scores, which decreased by 0.009 to 0.013 score yr⁻¹, offering evidence that resistance to lodging was also an important agronomic trait for selection by breeders (Table 3, Fig. 4). The genetic gain for lodging score improvement was equivalent to a lodging score unit of 1.0 between old

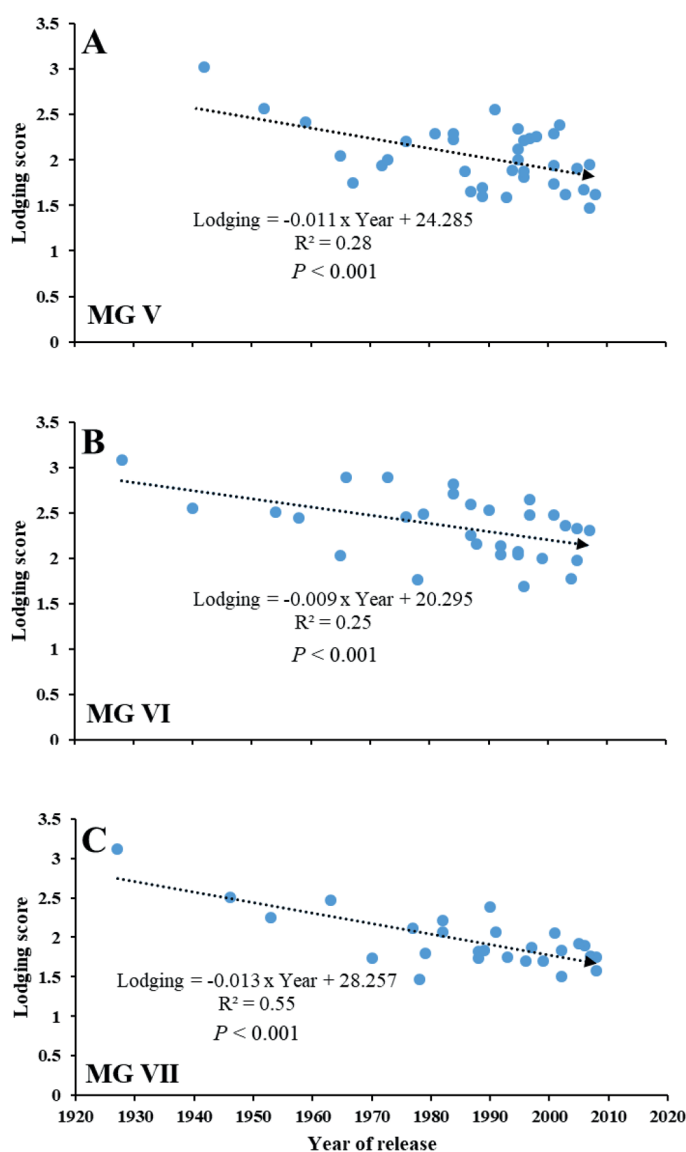


Fig. 4. Scatterplots and regressions depicting lodging improvement estimated for soybean cultivars in Maturity Groups (MGs) V through VII released from 1928 to 2008: (A) MG V ($n = 36$), (B) MG VI ($n = 30$), and (C) MG VII ($n = 27$). Estimates were based on best linear unbiased predictor (BLUP) trait values generated from replicated yield trials conducted in 27 total southern US year-location environments in 2010 to 2011, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII. The BLUP trait value for lodging score of each cultivar was regressed on the year of cultivar release to estimate annualized genetic yield gains made over eight decades for each MG.

and new cultivars across eight decades, a result similar to that observed by Rincker et al. (2014) between old and new MG I, II, and III cultivars (-0.012 to -0.018 score yr⁻¹) evaluated during the same time period; decreased lodging score was also observed by Luedders (1977), Voldeng et al. (1997), Ustun et al. (2001), Rogers et al. (2015), and de Felipe et al. (2016). Although changes in plant height (-0.032 to -0.118 cm yr⁻¹) were deemed not significant ($P > 0.05$) between the old and new MG V, VI,

and VII cultivars evaluated herein, the accumulated plant height reduction across 80 yr did amount to ~ 9.4 cm for newer MG V cultivars (-0.118 cm yr $^{-1}$) compared with older cultivars and ~ 5.0 cm for newer MG VIII cultivars (-0.061 cm yr $^{-1}$) compared with older cultivars (Table 3). This trend was observed in other genetic gain studies for MG 00 through VIII soybean cultivars (Boerma, 1979; Specht and Williams, 1984; Ustun et al., 2001; Wilcox, 2001; Rincker et al., 2014). Similar to plant height, no significant differences ($P > 0.05$) were found for days to maturity between the old and new cultivars within each MG, but a trend was observed whereby newer MG V and VII cultivars were faster to reach plant maturity than older cultivars across 80 yr by a cumulative 4 d for MG V (0.054 d yr $^{-1}$) and nearly 2 d for MG VII (0.022 d yr $^{-1}$) (Table 3). This result agrees with MG V maturity results reported by Ustun et al. (2001), yet contrasts with the trend that newer cultivars are slower to reach plant maturity in MGs II, III, and IV reported by Rincker et al. (2014) and no changes in plant maturities noted for MG IV to VI cultivars by Rogers et al. (2015). In the Mid-South, drought during the grain-filling stage (mid-July through mid-September) can be avoided by planting earlier-maturing cultivars, resulting in higher yields (Bowers, 1995; Heatherly and Spurlock, 1999). Therefore, cultivars that are faster to reach maturity in MG V may be a result of breeders integrating earlier maturing MG III and IV cultivars into their breeding programs. No significant differences were evident between old and new cultivars within each MG for 100-seed weight, which agrees with observations in genetic gain studies conducted for MG II to IV cultivars by Rincker et al. (2014) and for MG III to V cultivars by de Felipe et al. (2016).

Pearson correlation coefficients were generated to evaluate the associations between yield and the measured agronomic traits within each MG. The higher the correlation, the more likely that direct selection for yield will lead to indirect (i.e., positively or negatively correlated) selection for the trait. It is possible that in some cases, breeders may have directed selection at a trait through directional selection for yield. Across MGs, seed yield and lodging ($r = -0.51$ to -0.68) were the only two agronomic traits that were highly significantly correlated ($P < 0.001$) (Supplemental Table S2). For MG V, seed yield was also significantly ($P < 0.05$) correlated with plant height ($r = -0.41$). Lodging and plant height were significantly ($P < 0.01$) correlated for MG V ($r = 0.64$) and MG VI ($r = 0.48$), but not MG VII. Days to maturity was significantly ($P < 0.05$) correlated with both lodging and plant height in MG VI and VII.

Seed Composition and Quality

It is widely reported that seed protein concentration and seed yield share an inverse relationship (Burton, 1984;

Wilcox and Cavins, 1995; Cober and Voldeng, 1999). To date, no positive significant linear trends for seed protein have been recognized in any of the previously conducted genetic gain studies (Wilcox et al., 1979; Ustun et al., 2001; Rincker et al., 2014; Rogers et al., 2015; de Felipe et al., 2016). In the present study, seed protein concentration was unchanged for MG V but was significantly reduced for MG VI (-0.234 g kg $^{-1}$ yr $^{-1}$, $P < 0.01$) (Fig. 5B, Table 3), a reduction similar to that observed between new and old MG II to IV cultivars by Rincker et al. (2014). Additionally, no change in seed protein concentration was observed for MG VII (Fig. 5C). Concurrent selection that may have occurred for both high protein and high yield offers one explanation for the fact that seed protein levels were maintained with gains in yield for MGs V and VII. Conversely, no significant changes were observed for seed oil concentration across 80 yr for any of the MGs studied (Table 3). This result is in contrast with increases in seed oil concentrations that were noticed in genetic gain studies conducted by Wilcox et al. (1979), Ustun et al. (2001), Rincker et al. (2014), and Rogers et al. (2015). Like protein and yield, protein and oil are inversely related (Wilson, 2004), so reductions in protein and concordant increases in oil in these respected studies were expected. Seed quality score (scale score was reduced) only significantly improved ($P < 0.005$) across time in MG V (Table 3).

Pearson correlation coefficients showed that yield was not correlated with any of the measured quality traits across MGs. For MG V, seed yield was only significantly correlated ($P < 0.001$) with seed quality. For MG VI, seed yield was significantly ($P < 0.01$) negatively correlated with protein ($r = -0.53$) and significantly ($P < 0.05$) positively correlated with oil ($r = 0.39$). For MG VII, the correlation between seed yield and seed protein ($r = -0.34$) was noteworthy but not significant ($P = 0.08$) (Supplemental Table S2).

Yield Stability Responses

A yield stability analysis was conducted using the Finlay–Wilkinson method to assess whether there were yield stability changes between the old and new cultivars within each MG when measured across the 80-yr period (Finlay and Wilkinson, 1963). As listed in Table 4 and displayed in Fig. 6, higher regression coefficients were observed for newer cultivars when compared with older cultivars within each MG. Furthermore, highly significant positive linear trends ($P < 0.001$) were observed within MGs when the regression coefficient of each cultivar was regressed on the year of release, and these trends were almost identical among the three evaluated MGs (0.005 – 0.006 yr $^{-1}$) (Fig. 6). The regression coefficients for the cultivars ranged from 0.74 to 1.24 in MG V, from 0.62 to 1.41 in MG VI and from 0.51 to 1.38 in MG VII (Table 4). When the regression coefficient (b) = 1.0, the yield stability of that cultivar is deemed to be average, or similar to

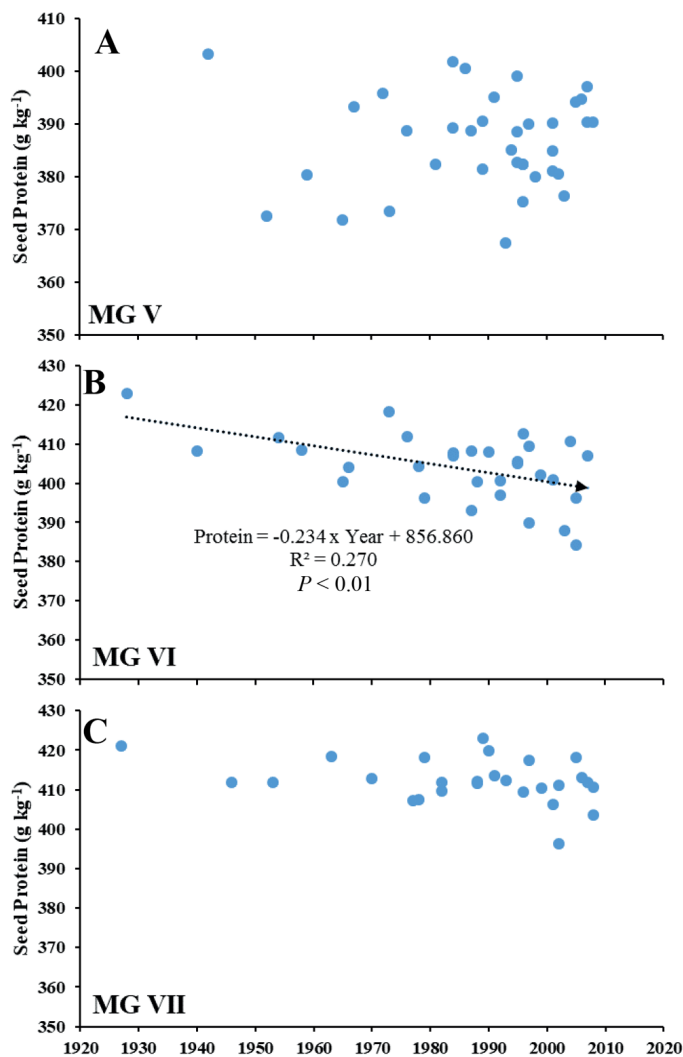


Fig. 5. Scatterplots and regressions depicting genetic changes to seed protein concentration estimated for soybean cultivars in Maturity Groups (MGs) V through VII released from 1928 to 2008: (A) MG V ($n = 36$), (B) MG VI ($n = 30$), and (C) MG VII ($n = 27$). Estimates were based on best linear unbiased predictor (BLUP) trait values generated from replicated yield trials conducted in 27 total southern US year-location environments in 2010 to 2011, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII. The BLUP trait value for seed protein concentration of each cultivar was regressed on the year of cultivar release to estimate annualized genetic changes to seed protein concentration made over eight decades for each MG.

the mean stability yield response of the other cultivars evaluated in the same environments (Tollenaar and Lee, 2002; Bernardo, 2010; Rincker et al., 2014). Therefore, the higher stability coefficients of the newer cultivars than the older cultivars within each MG across the evaluated environments indicated that they might have had a greater yield response in favorable production environments compared with the older cultivars. This result is consistent with the observation from Rincker et al. (2014). The newer cultivars with $b > 1.0$ indicated a rather low stability, whereas the older cultivars with $b < 1.0$ suggested a rather high stability (Tollenaar and Lee, 2002). The number of cultivars with regression coefficients

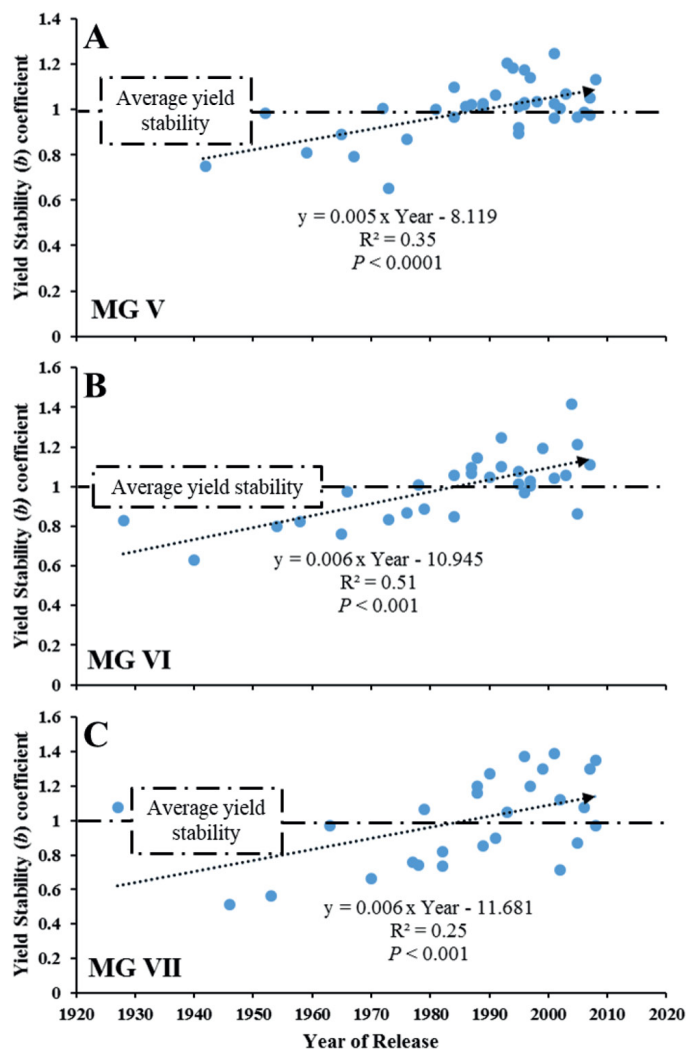


Fig. 6. Scatterplots depicting regression of yield stability with year of release for soybean cultivars released from 1928 to 2008: (A) Maturity Group (MG) V ($n = 36$), (B) MG VI ($n = 30$), and (C) MG VII ($n = 27$). Estimates were based on regression coefficient (b) values generated from the mean of cultivar yield using the Finlay-Wilkinson model (Finlay and Wilkinson, 1963) when evaluated in 27 total southern US year-location environments in 2010 to 2011, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII.

(b) at or near 1.0 was highest in MG V (Fig. 6A, Table 4), in comparison with the regression coefficients calculated for MGs VI (Fig. 6B, Table 4) and VII (Fig. 6C, Table 4). The cultivars '5601T' ($b = 1.25$, MG V) '96M20' ($b = 1.41$, MG VI), and '97B52' ($b = 1.38$, MG VII) had the highest regression coefficients within their respective MGs, indicating that they were less stable, but conversely clearly more responsive to modern favorable production environments (Table 4). In general, the magnitude changes in yield stability regression coefficients observed across time in the present study were very similar to yield stability results observed by Rincker et al. (2014) for each group of MG II to IV soybean cultivars that were estimated at 0.005 yr^{-1} and evaluated during the same 80-yr time period.

Table 4. Yield stability regression coefficients estimated for soybean cultivars selected from Maturity Group (MG) V through VII released from 1928 to 2008. Estimates were based on regression coefficient (*b*) values generated from the mean of cultivar yield using the Finlay–Wilkinson model (Finlay and Wilkinson, 1963) when evaluated in 27 total southern US year-location environments from 2010 to 2011.

MG	Cultivar	Accession no.	Year of Release	<i>b</i> †	SE	<i>P</i>
V	S-100	PI 548488	1942	0.75	0.07	<0.0001
	Dorman	PI 548653	1952	0.99	0.10	<0.0001
	Hill	PI 548654	1959	0.81	0.07	<0.0001
	Dare	PI 548987	1965	0.89	0.09	<0.0001
	York	PI 553038	1967	0.80	0.11	<0.0001
	Essex	PI 548667	1972	1.00	0.09	<0.0001
	Forrest	PI 548655	1973	0.65	0.10	<0.0001
	McNair 500	PI 556506	1976	0.87	0.09	<0.0001
	MON033	NA‡	1981§	1.00	0.08	<0.0001
	9592	PI 556915	1984	0.97	0.08	<0.0001
	Coker 485	PI 556743	1984	1.10	0.09	<0.0001
	MON043	NA	1986§	1.01	0.07	<0.0001
	Hutcheson	PI 518664	1987	1.02	0.08	<0.0001
	Coker 6955	PI 556953	1989	1.03	0.07	<0.0001
	MON021	PI 556861	1989	1.01	0.07	<0.0001
	S59-60	NA	1991	1.06	0.08	<0.0001
	Holladay	PI 572239	1993	1.20	0.10	<0.0001
	Clifford	PI 596414	1994	1.18	0.07	<0.0001
	MON034	NA	1995§	0.89	0.08	<0.0001
	MON035	NA	1995§	0.92	0.09	<0.0001
	MON036	NA	1995	1.01	0.07	<0.0001
	9594	NA	1996	1.18	0.11	<0.0001
	MON022	NA	1996	1.03	0.10	<0.0001
	Graham	PI 594922	1996	1.02	0.09	<0.0001
	S59-V6	NA	1997	1.14	0.10	<0.0001
	MON023	NA	1998	1.03	0.10	<0.0001
	MON028	NA	2001	1.03	0.08	<0.0001
	MON040	NA	2001	0.96	0.10	<0.0001
	5601T	PI 630984	2001	1.25	0.06	<0.0001
	S56-D7	NA	2002	1.01	0.08	<0.0001
	MON027	NA	2003	1.07	0.11	<0.0001
	95M50	NA	2005	0.97	0.06	<0.0001
	Jake	PI 643912	2006	0.99	0.09	<0.0001
	S52-F2	NA	2007	1.05	0.07	<0.0001
	Osage	PI 648270	2007	0.97	0.13	<0.0001
	95Y40	NA	2008	1.13	0.10	<0.0001
VI	Arksoy	PI 548438	1928	0.83	0.08	<0.0001
	Ogden	PI 548477	1940	0.63	0.11	<0.0001
	Lee	PI 548656	1954	0.80	0.10	<0.0001
	Hood	PI 548980	1958	0.82	0.09	<0.0001
	Pickett	PI 548988	1965	0.76	0.10	<0.0001
	Davis	PI 553039	1966	0.97	0.11	<0.0001
	Tracy	PI 548983	1973	0.83	0.13	<0.0001
	Centennial	PI 548975	1976	0.87	0.10	<0.0001
	Coker 136	PI 628819	1978	1.01	0.10	<0.0001
	Coker156	PI 556564	1979	0.89	0.10	<0.0001
	Leflore	PI 548981	1984	0.85	0.10	<0.0001
	Young	PI 508266	1984	1.05	0.11	<0.0001
	Coker 686	PI 556827	1987	1.07	0.08	<0.0001
	MON026	PI 527704	1987	1.10	0.09	<0.0001
	MON024	PI 556858	1988	1.14	0.11	<0.0001
	Brim	PI 548986	1990	1.05	0.13	<0.0001
	9692	NA	1992	1.10	0.12	<0.0001
	S65-50	NA	1992	1.25	0.15	<0.0001

Table 4. Continued.

MG	Cultivar	Accession no.	Year of Release	$b\ddagger$	SE	P
MG	MON025	NA	1995	1.08	0.13	<0.0001
	Dillon	NA	1995	1.01	0.11	<0.0001
	S61-Q2	NA	1996	0.97	0.09	<0.0001
	Boggs	PI 602597	1997	1.00	0.12	<0.0001
	Musen	PI 599333	1997	1.03	0.09	<0.0001
	96B21	NA	1999	1.19	0.14	<0.0001
	NC Roy	PI 617045	2001	1.04	0.16	<0.0001
	MON032	NA	2003	1.06	0.13	<0.0001
	96M20	NA	2004	1.42	0.13	<0.0001
	MON030	NA	2005	0.86	0.16	<0.0001
	S65-M3	NA	2005	1.22	0.08	<0.0001
	S62-66	NA	2007	1.11	0.10	<0.0001
VII	CNS	PI 548445	1927	1.08	0.17	<0.0001
	Roanoke	PI 548485	1946	0.51	0.14	0.0009
	Jackson	PI 548657	1953	0.56	0.12	<0.0001
	Bragg	PI 548660	1963	0.97	0.12	<0.0001
	Ransom	PI 548989	1970	0.66	0.12	<0.0001
	GaSoy17	PI 553046	1977	0.76	0.16	<0.0001
	Coker 237	PI 556536	1978	0.74	0.14	<0.0001
	Braxton	PI 548659	1979	1.07	0.12	<0.0001
	Coker 317	PI 556623	1982	0.73	0.14	<0.0001
	MON039	NA	1982§	0.82	0.12	<0.0001
	Coker 6847	PI 556862	1988	1.20	0.10	<0.0001
	Thomas	PI 522236	1988	1.16	0.12	<0.0001
	Stonewall	PI 531068	1989	0.85	0.13	<0.0001
	Hagood	PI 555453	1990	1.27	0.12	<0.0001
	Cook	PI 553045	1991	0.90	0.16	<0.0001
	S75-55	NA	1993	1.05	0.15	<0.0001
	Benning	PI 595645	1996	1.37	0.14	<0.0001
	S73-Z5	NA	1997	1.20	0.16	<0.0001
	MON044	NA	1999	1.30	0.13	<0.0001
	97B52	NA	2001	1.39	0.14	<0.0001
	S76-L9	NA	2002	1.12	0.16	<0.0001
	NC-Raleigh	PI 641156	2002	0.71	0.30	0.0236
	Woodruff	NA	2005	0.87	0.15	<0.0001
	97M50	NA	2006	1.08	0.11	<0.0001
	N7002	PI 647085	2007	1.30	0.13	<0.0001
	MON031	NA	2008	1.35	0.17	<0.0001
	S74-W6	NA	2008	0.97	0.21	<0.0001

† Significant at the $P < 0.001$ probability level.

‡ NA, not available due to company policy.

§ Denotes the first year cultivar was found in variety testing reports.

CONCLUSIONS

Across an 80-yr period from 1930 to 2010, great progress was made for genetic gains of seed yield for MG V, VI, and VII soybeans in the United States, which were estimated at 17.6, 13.5 and 10.3 kg ha⁻¹ yr⁻¹, respectively (Fig. 2). Averaged across MGs V to VII, the annualized rate of genetic yield improvement was 13.7 g kg⁻¹ yr⁻¹ (Fig. 3). For agronomic traits, resistance to lodging improved by a lodging scale score of ~1.0 across eight decades, indicating that it was a trait selected on by breeders. Additionally, resistance to lodging was the only trait that was highly significantly ($P < 0.01$) correlated with yield in MG V

($r = -0.60$), VI ($r = -0.51$), and VII ($r = -0.68$). Cultivar height decreased across time within each MG and newer cultivars matured earlier than older cultivars in MG VI and VII, although maturity changes were not significant for the cultivars evaluated within each of the MGs.

Given the inverse relationship that exists between seed yield and seed protein, genetic yield gains made during the past 80 yr came at the direct expense of protein, which significantly ($P < 0.05$) declined by 12.2 g kg⁻¹ yr⁻¹ during the past eight decades across MGs V to VII (Fig. 7). Moreover, seed yield and seed protein were significantly negatively correlated across MGs in the present study ($P <$

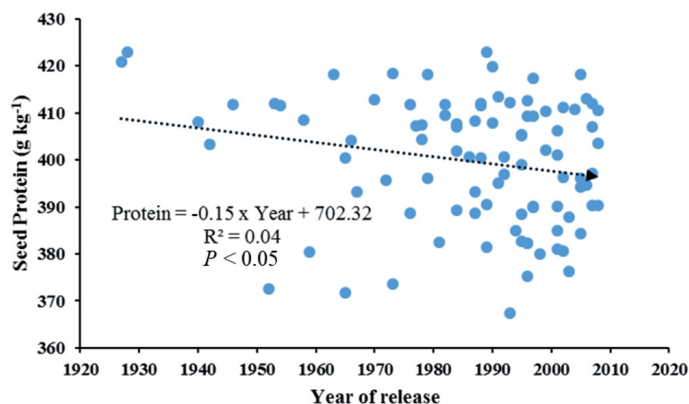


Fig. 7. Scatterplot depicting genetic changes for seed protein estimated for all 93 soybean cultivars selected from Maturity Group (MG) V, VI, and VII and released from 1928 to 2008. The estimates were based on best linear unbiased predictor (BLUP) trait values generated from replicated yield trials conducted in 27 total southern US environments from 2010 to 2011, which amounted to 12 environments for MG V, six environments for MG VI, and nine environments for MG VII.

0.0001, $r = -0.52$). No significant changes were detected for seed oil concentration. In light of the seed protein concentration reductions for North American MGs II to VII, moving forward, soybean breeders will have to be vigilant in addressing this concern by concurrently selecting for high yield and high protein.

Supplemental Material

Supplemental material is available online for this article.

Conflict of Interest

The authors declare that there is no conflict of interest.

Acknowledgments

This work was supported by funding from the United Soybean Board.

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